

## Experiment 7

### End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

|                     |                                               |             |
|---------------------|-----------------------------------------------|-------------|
| Degree & Branch     | B.Tech Artificial Intelligence & Data Science | Semester V  |
| Subject Code & Name | CS3807 – Deep Learning Laboratory             | AY: 2026–27 |

## 1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

## 2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

## 3. Dataset and Experimental Setup

### Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

[https://www.tensorflow.org/api\\_docs/python/tf/keras/datasets/mnist](https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist)

### Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where  $x$  is the input image and  $\hat{x}$  is its reconstruction.

## 4. Expected Input and Output

### Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

### Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image  
Reconstructed image  
Reconstruction error

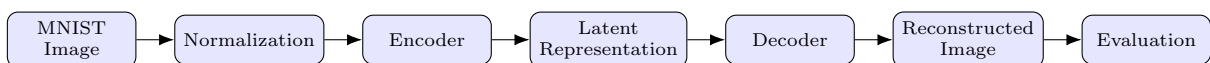
For the denoising model, display:

Original clean image  
Noisy image  
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

## 5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

## 6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

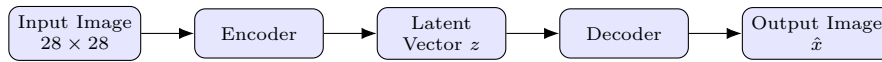
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

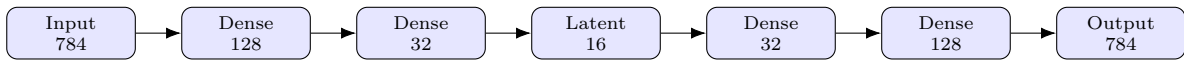
## 7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

**Suggested configuration:**

| Parameter        | Value                |
|------------------|----------------------|
| Latent dimension | 16                   |
| Optimizer        | Adam                 |
| Learning rate    | $10^{-3}$            |
| Batch size       | 128                  |
| Epochs           | 20                   |
| Loss             | Binary Cross-Entropy |

## 8. Required Analysis of the Basic Autoencoder

### Plot 1: Original versus Reconstructed Images

The trained fully connected autoencoder was evaluated on ten previously unseen test images. The original and reconstructed images are shown below.

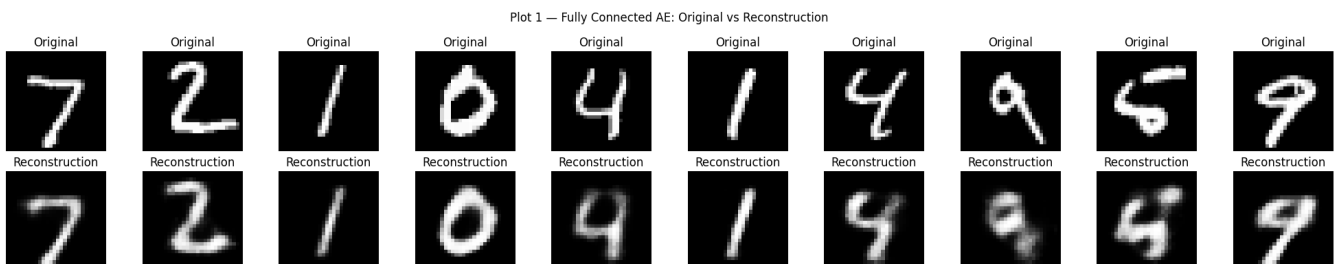


Figure 1: Original and reconstructed images using the fully connected autoencoder.

**Inference:** The reconstructed digits keep the main shape and identity, showing that the 16-dimensional latent space captures important information. However, the images are more smooth and some thin strokes and small details are lost. Overall, the model learns the basic digit structure but loses some fine details due to compression.

**Plot 2: Training and Validation Reconstruction Loss**

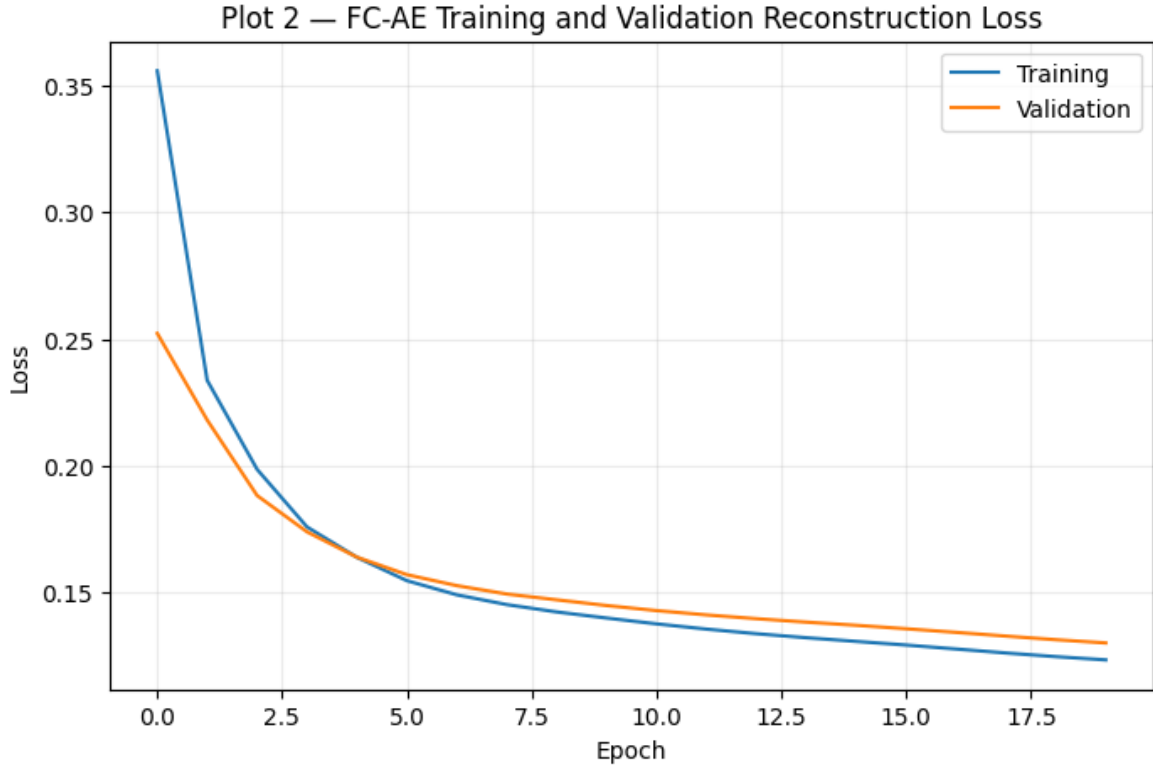


Figure 2: Training and validation reconstruction loss of the fully connected autoencoder.

**Inference:** Both training and validation loss decreases steadily over 20 epochs, reaching about 0.1234 and 0.1301. The curves stays close, showing good learning with no clear sign of overfitting.

## 9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

### Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

### Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

### Structural Similarity

Calculate SSIM between the original and reconstructed images.

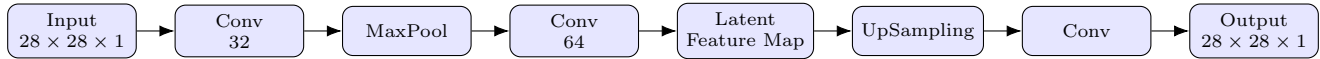
| Metric    | Value    |
|-----------|----------|
| Test MSE  | 0.022020 |
| Test MAE  | 0.058438 |
| Mean SSIM | 0.737232 |

**Inference:**The FC-AE has relatively low MSE and MAE, while the SSIM of 0.737232 shows that the main digit structure is preserved. The digits are recognizable, but some fine details are lost, so it is less effective than the convolutional model.

## 10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```

Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
  
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

## 11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

| Model              | MSE      | MAE      | SSIM     | Parameters |
|--------------------|----------|----------|----------|------------|
| Fully Connected AE | 0.022020 | 0.058438 | 0.737232 | 211040     |
| Convolutional AE   | 0.002623 | 0.015045 | 0.973955 | 74497      |

Plot 3: Reconstruction Comparison

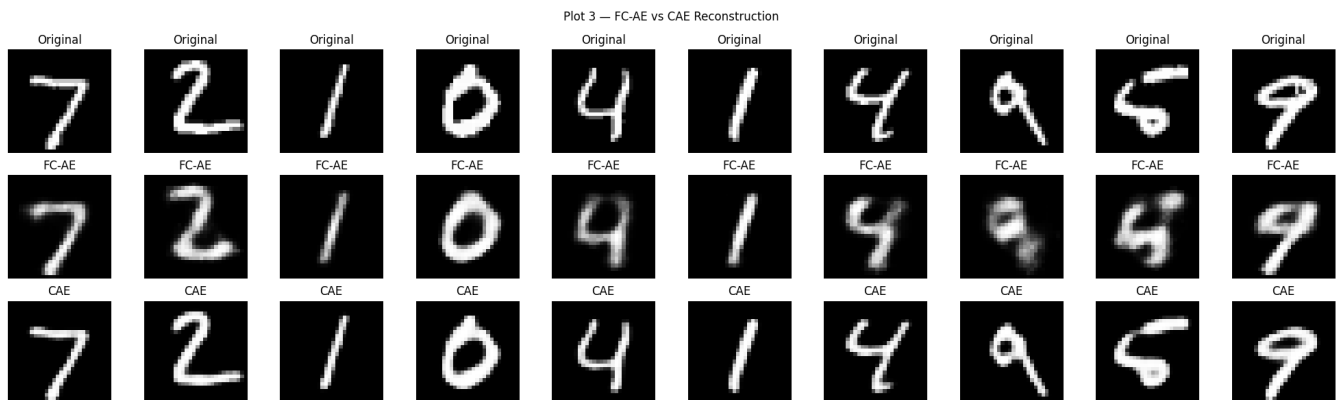


Figure 3: Comparison of fully connected and convolutional autoencoder reconstructions.

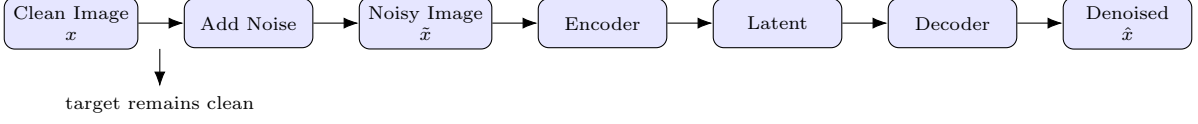
**Inference:**The CAE gives more sharper and better reconstructions than the FC-AE, with lower MSE and higher SSIM. Convolutional layers preserves spatial information better, making it more usefull for image reconstruction.

## 12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image. The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where  $\tilde{x}$  is a noisy version of the clean image  $x$ .



## 13. Adding Image Noise

Use at least two corruption types.

### Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range  $[0, 1]$ .

### Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

## 14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

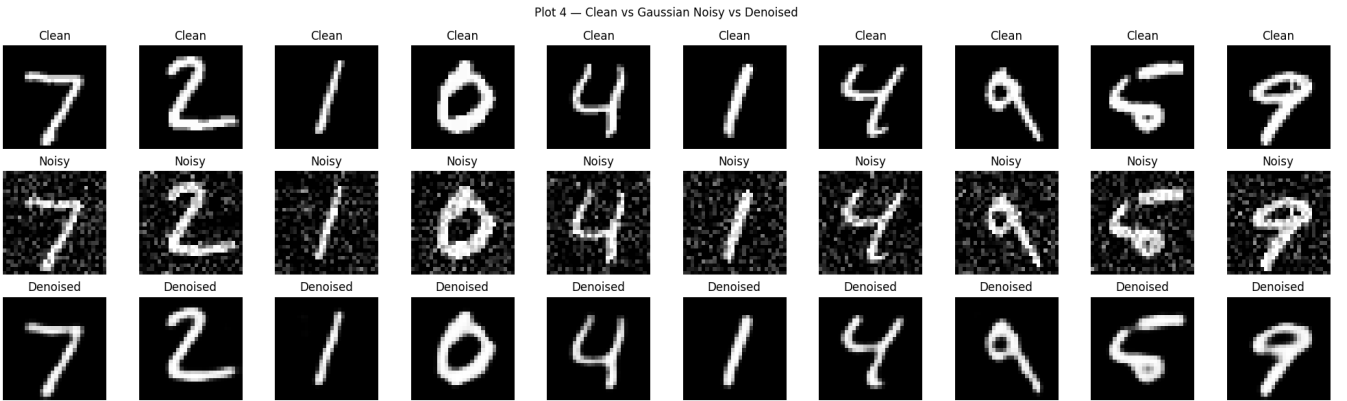


Figure 4: Clean, Gaussian-noisy and denoised test images for  $\sigma = 0.2$ .

**Inference:** The Gaussian noise affects the digit strokes and background, but the denoising CAE removes most of the noise. The reconstructed images keep the main digit shape, although some small details become smooth. Overall, the model can recover the important structure even from noisy images.

Plot 5: Noise Level vs. Reconstruction Quality

| Noise Level ( $\sigma$ ) | MSE      | MAE      | SSIM     |
|--------------------------|----------|----------|----------|
| 0.1                      | 0.003778 | 0.018272 | 0.956450 |
| 0.2                      | 0.004944 | 0.021993 | 0.941735 |
| 0.3                      | 0.007549 | 0.029805 | 0.878786 |

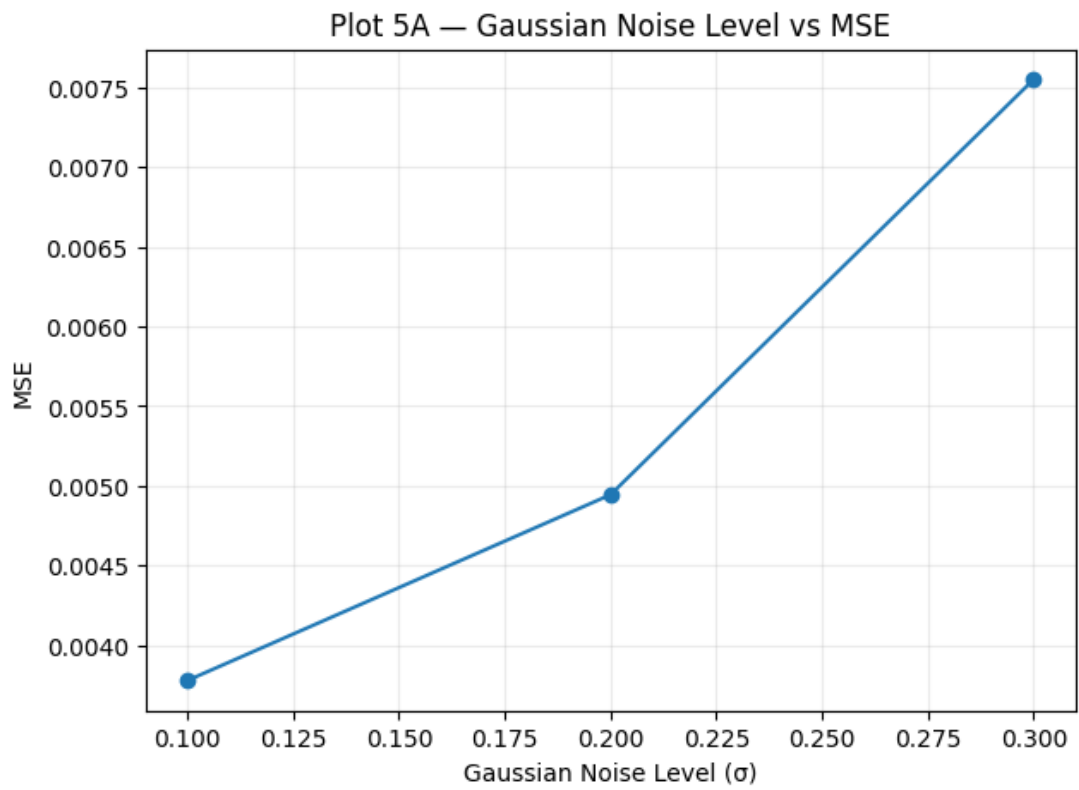


Figure 5: Gaussian noise level versus MSE.

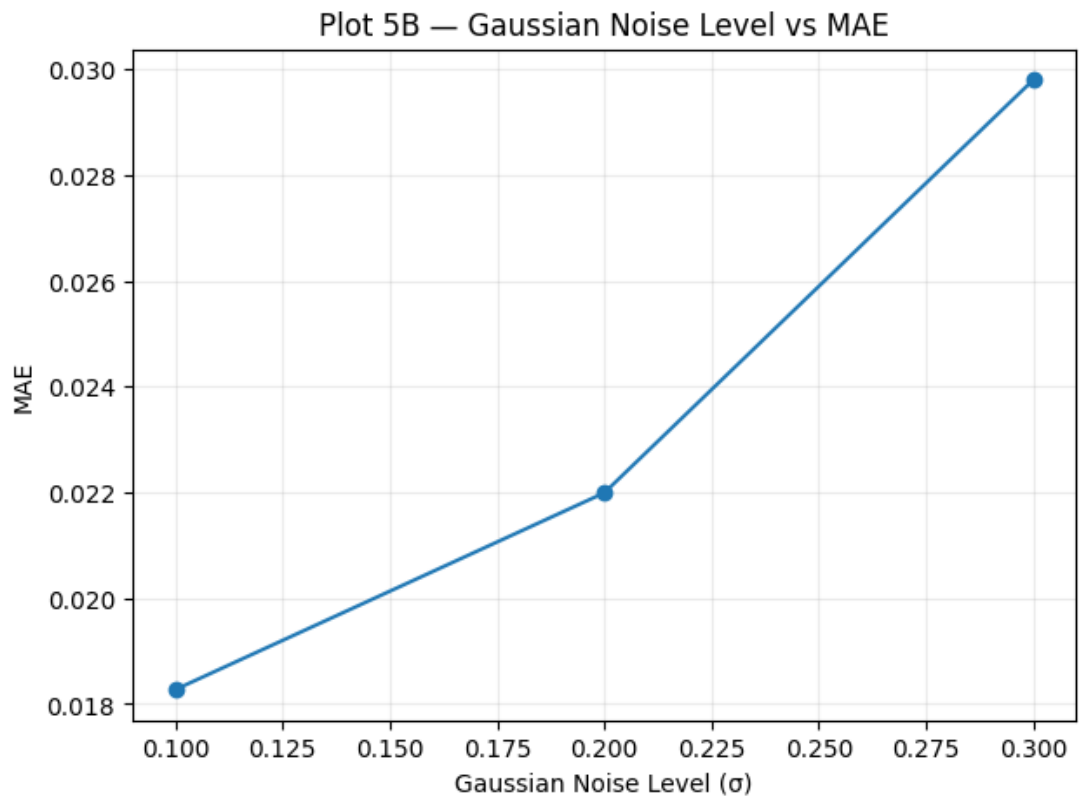


Figure 6: Gaussian noise level versus MAE.

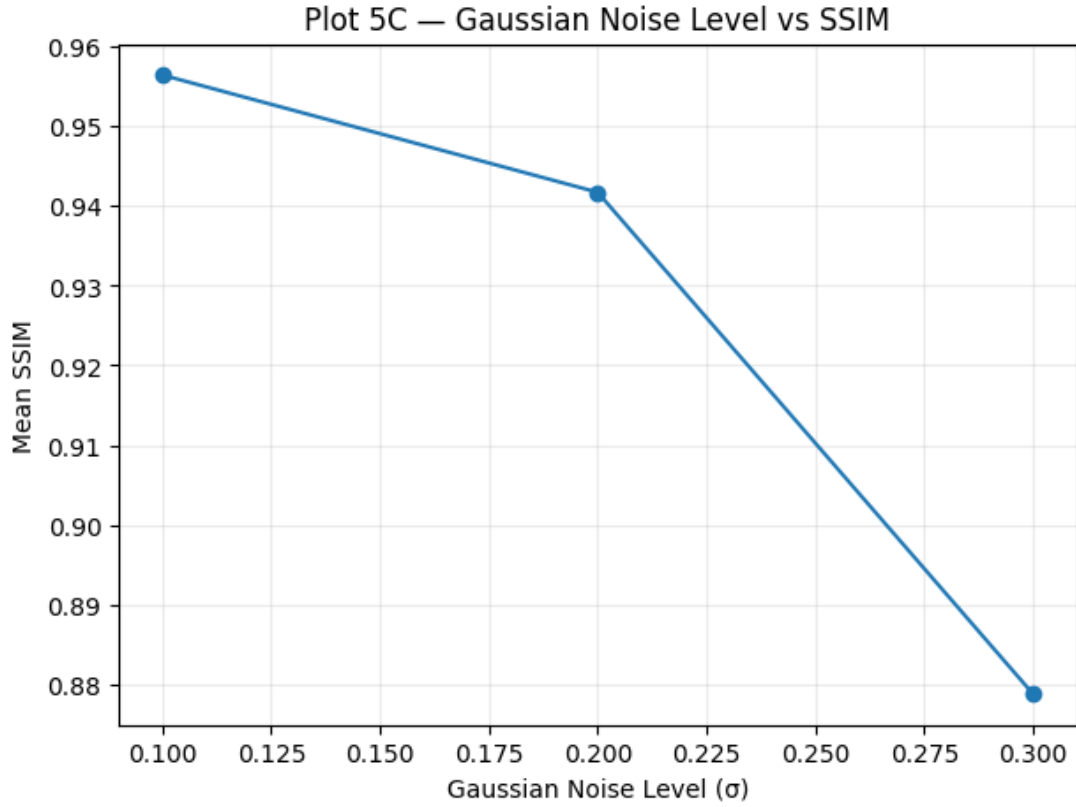


Figure 7: Gaussian noise level versus SSIM.

**Inference:** As the noise increases from  $\sigma = 0.1$  to  $0.3$ , MSE and MAE increase while SSIM decreases. This is because more noise makes the image harder to reconstruct, but the model still manages to recover the main digit structure.

## 15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

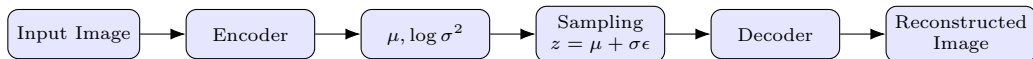
$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



## 16. VAE Loss Function

The VAE objective consists of two components.

### Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

## KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

## 17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

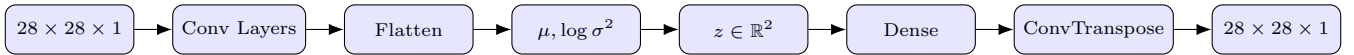
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



## 18. Exploring the VAE Latent Space

Since the latent dimension is two, the encoded test images can be visualized in a two-dimensional plane.

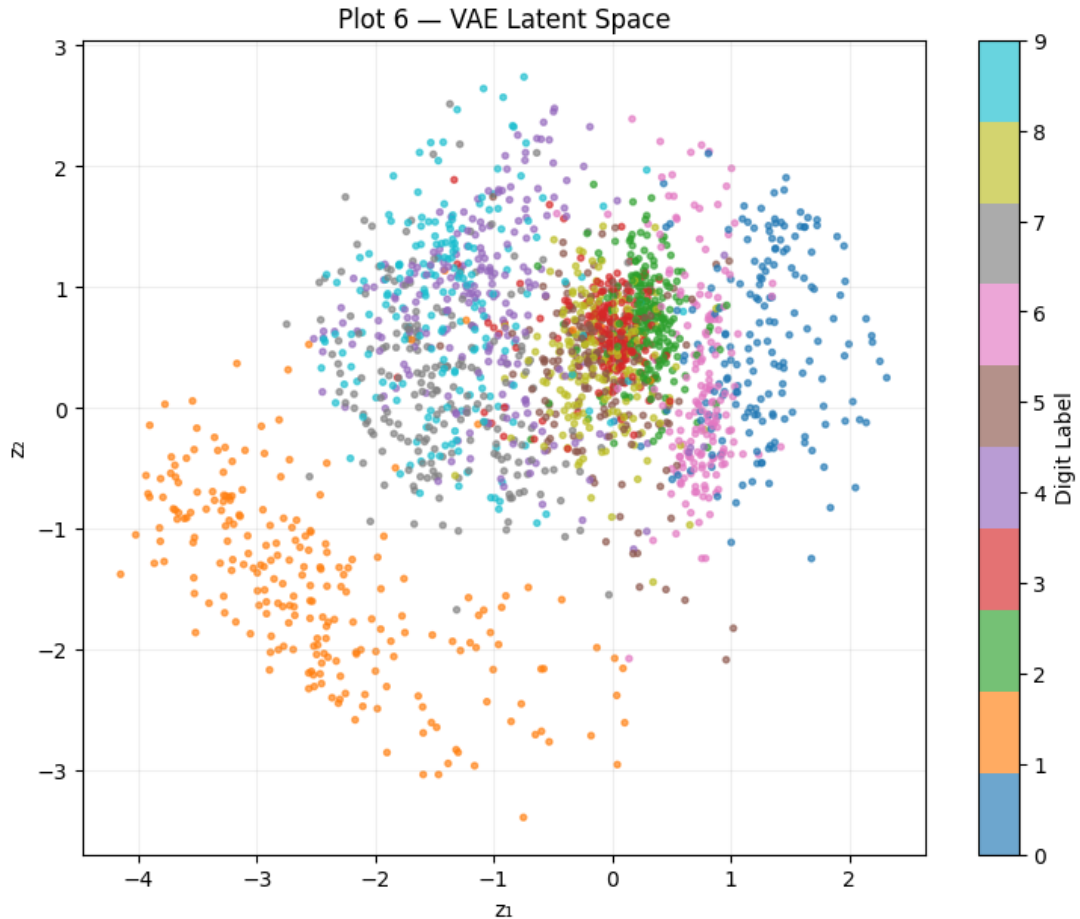


Figure 8: Two-dimensional VAE latent-space visualization. Labels are used only for visualization.

**Inference:** The latent plot shows that the VAE groups many handwritten digits into different regions, but some classes are still overlapping. This is because some digits have similar shapes and strokes. The spread of points also shows that the VAE uses a continuous latent space instead of one fixed point for each digit.

## 19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Pass the sampled latent vectors through the decoder.

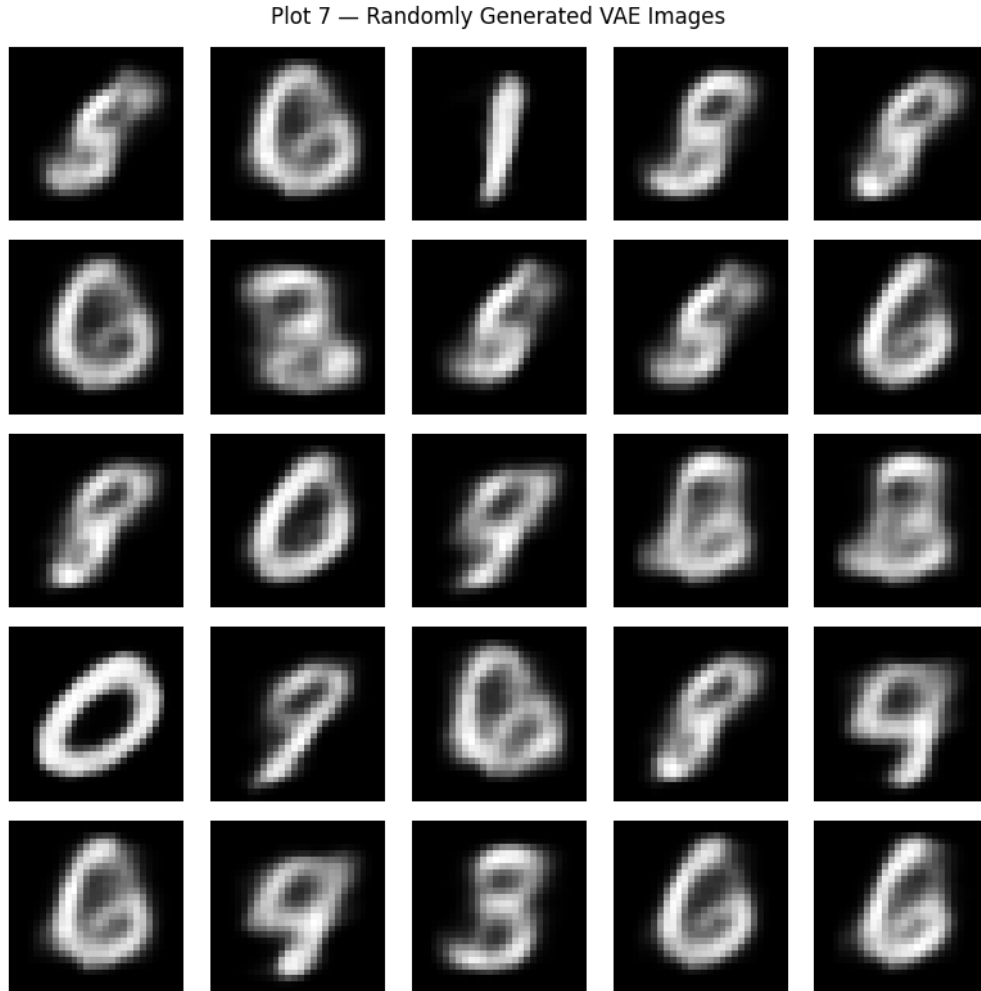


Figure 9: Twenty-five randomly generated images from the VAE prior.

**Inference:** The generated images mostly look like handwritten digits, and some are clearly recognizable like 5, 6, 8 and 9. The images are a little soft and blurry, but they still show different stroke shapes and thickness. Some samples are not very clear, but overall the VAE is able to generate new digit images.

## 20. Latent Space Interpolation

Select two latent vectors  $z_A$  and  $z_B$  and construct

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

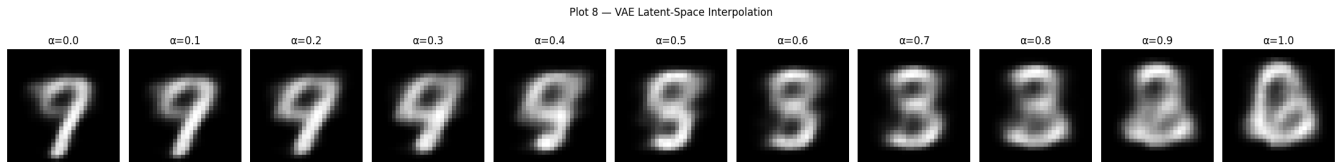


Figure 10: VAE latent-space interpolation for  $\alpha = 0, 0.1, \dots, 1.0$ .

**Inference:** The decoded images changes gradually as  $\alpha$  moves from 0 to 1. The sequence starts with a recognizable 9 and slowly changes its stroke structure until it looks like a 6. The middle images are also clear and does not change suddenly, showing that the VAE latent space can create smooth and meaningful transitions.

## 21. Reconstruction Comparison Across Models

Compare the three reconstruction-based models:

| Model                            | MSE      | MAE      | SSIM     | Parameters |
|----------------------------------|----------|----------|----------|------------|
| Fully Connected AE               | 0.022020 | 0.058438 | 0.737232 | 211040     |
| Convolutional AE                 | 0.002623 | 0.015045 | 0.973955 | 74497      |
| Denoising CAE ( $\sigma = 0.2$ ) | 0.004915 | 0.021914 | 0.942025 | 74497      |

For the VAE, report its reconstruction loss separately because its total objective also contains the KL-divergence term.

| VAE Metric          | Value      |
|---------------------|------------|
| Reconstruction Loss | 152.007645 |
| KL Loss             | 5.442253   |
| Total Loss          | 157.449898 |
| Test MSE            | 0.045195   |
| Test MAE            | 0.107254   |
| Mean SSIM           | 0.468692   |

**Inference:** The CAE gives the best clean-image reconstruction, while the denoising CAE performs slightly lower because it removes noise first. The FC-AE loses more detail, while the VAE sacrifices some reconstruction quality to maintain a useful probabilistic latent space.

## 22. Required Plots

The final report contains all ten required visual analyses specified in the experiment. The figures are presented in their respective sections above: original-versus-reconstructed images, FC-AE training/validation loss, FC-AE versus CAE reconstruction, clean/noisy/denoised images, Gaussian noise-level metrics, VAE latent space, generated VAE samples, latent interpolation, VAE reconstruction-loss curves, and the reconstruction-error distribution.

The VAE training and validation reconstruction-loss curves are shown below.

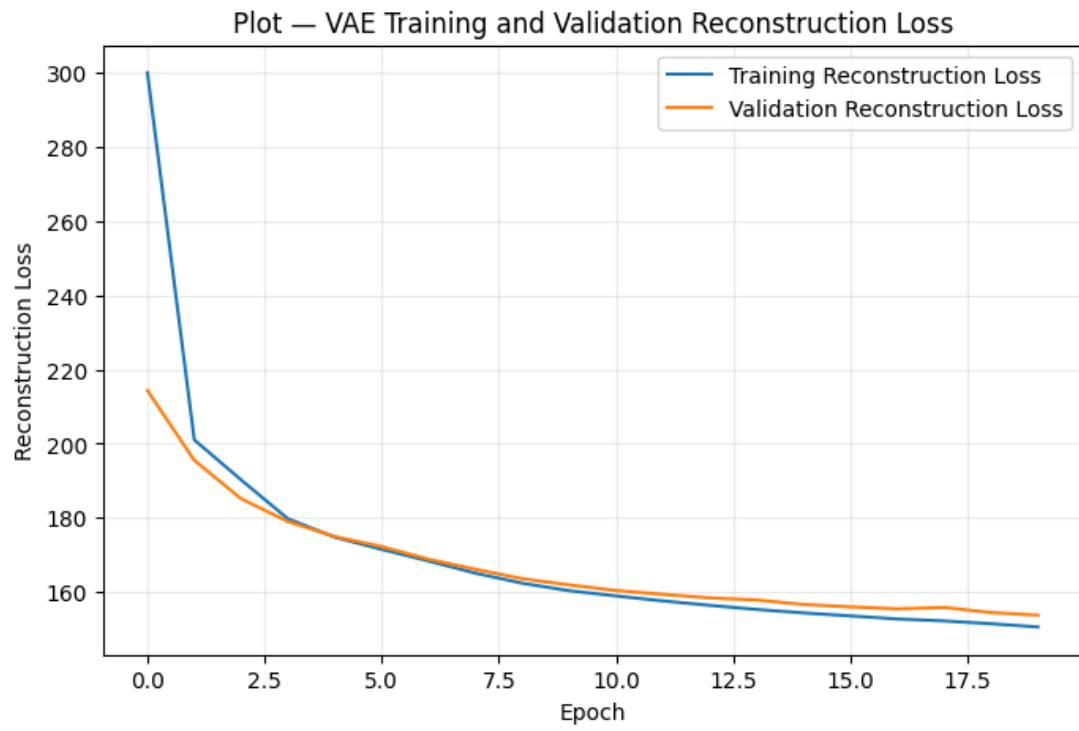


Figure 11: VAE training and validation reconstruction loss.

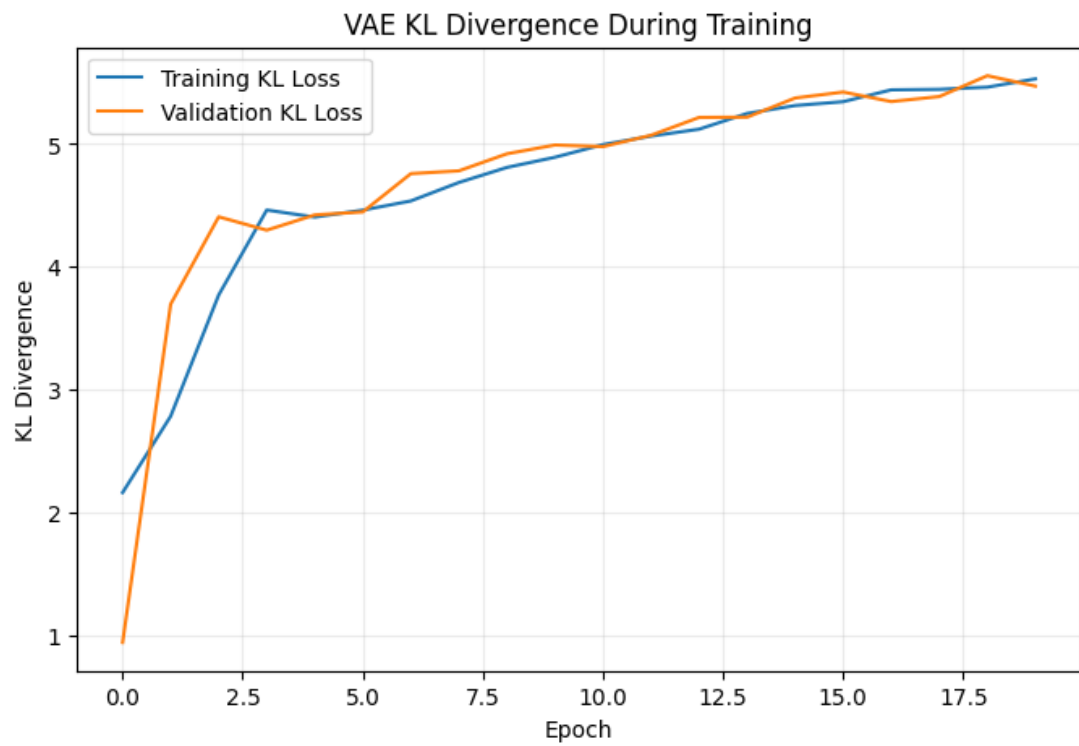


Figure 12: VAE training and validation KL loss.

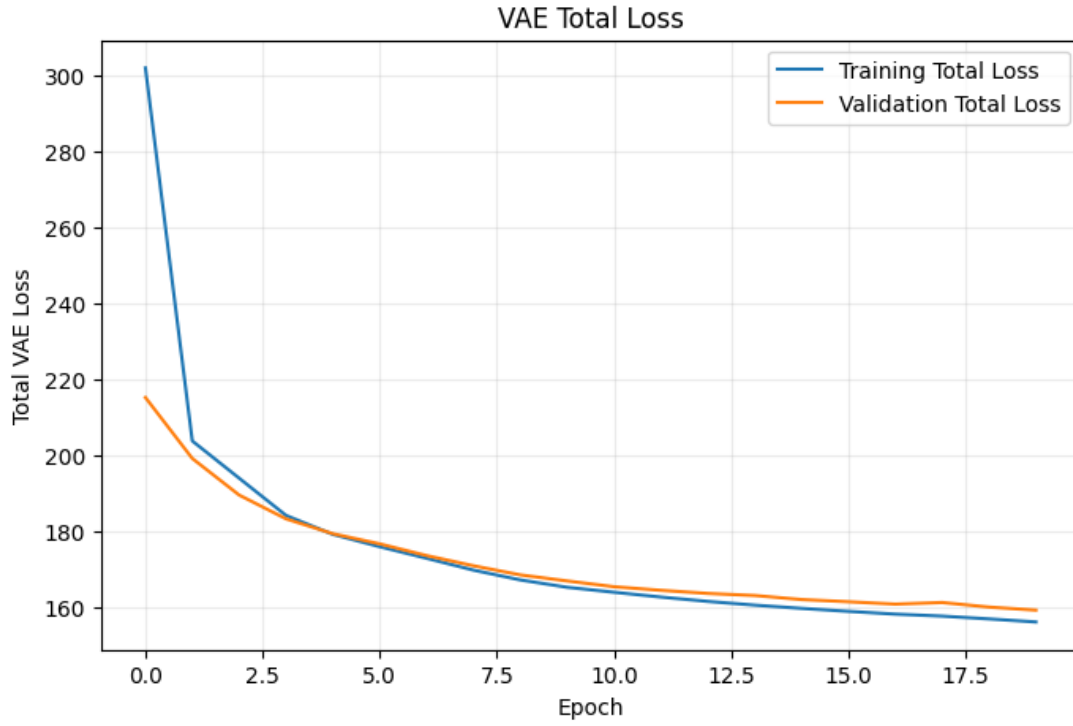


Figure 13: VAE training and validation total loss.

**Inference:** The VAE reconstruction loss decreases during training, showing that the model is getting better at reconstructing images. The training and validation curves remain similar, while the KL loss regularizes the latent space with no clear sign of overfitting.

## 23. Reconstruction Error Distribution

Calculate the per-image reconstruction error:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

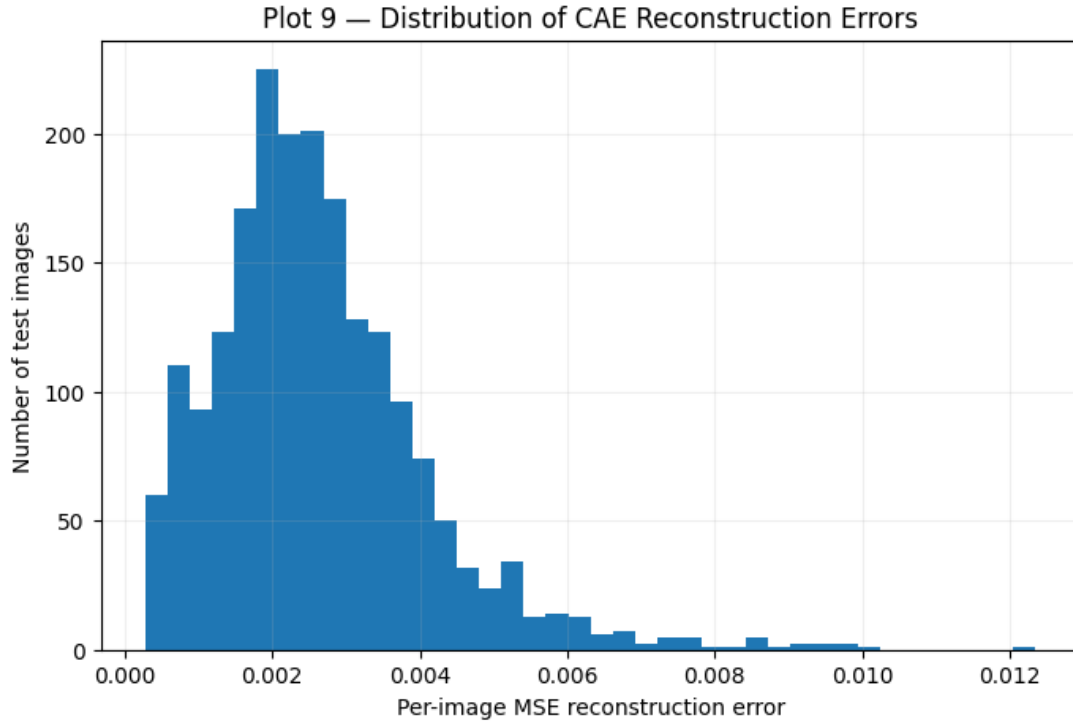


Figure 14: Distribution of per-image reconstruction errors for the CAE on the test set.

The recorded statistics were:

| Statistic     | Value    |
|---------------|----------|
| Minimum error | 0.000278 |
| Maximum error | 0.012352 |
| Mean error    | 0.002623 |
| Median error  | 0.002424 |

**Inference:** The reconstruction-error distribution is concentrated near the lower end of the observed range, with a mean of 0.002623 and a median of 0.002424. This indicates that most test images are reconstructed with relatively small pixel-wise error. A smaller number of images form the high-error tail, which identifies samples whose stroke structure or writing style is less well represented by the learned latent features. Therefore, the histogram is useful not only for measuring average performance but also for locating individual cases where visual reconstruction deserves closer inspection.

## 24. Required High-Error Image Analysis

The five test images with the largest CAE reconstruction errors were identified from the test set.

| Sample | Test Index | Digit Label | Reconstruction Error |
|--------|------------|-------------|----------------------|
| 1      | 895        | 0           | 0.012352             |
| 2      | 1395       | 2           | 0.009989             |
| 3      | 1982       | 6           | 0.009773             |
| 4      | 431        | 8           | 0.009709             |
| 5      | 1526       | 0           | 0.009611             |

Five Highest-Error CAE Test Images

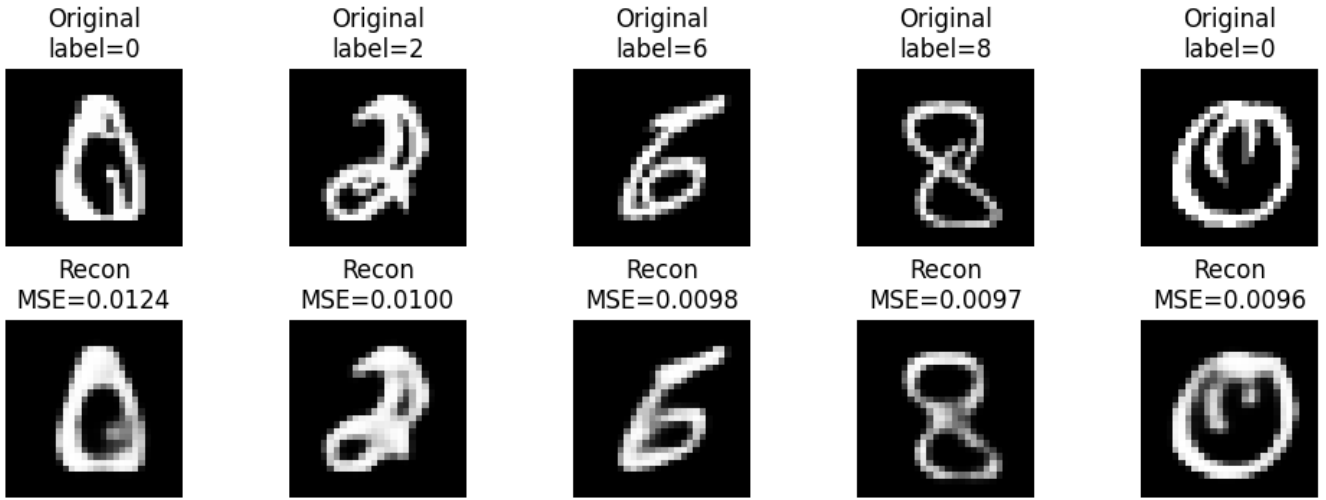
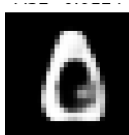


Figure 15: Five highest-error test images and their CAE reconstructions.

| Sample | Original                                                                            | Reconstruction                                                                      |
|--------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1      |    |    |
| 2      |   |   |
| 3      |  |  |
| 4      |  |  |
| 5      |  |  |

**Inference:** These five samples are difficult mainly because their handwriting contains stroke shapes, thicknesses or proportions that differ from the more common patterns learned by the CAE. The reconstructions still preserve the identity of the digits, but the stroke boundaries are smoother and some internal or curved details are simplified. The high-error images therefore illustrate an important point: a larger numerical reconstruction error does not necessarily mean that the digit becomes unrecognizable; it can also arise from relatively small but spatially distributed differences between the original handwriting and the model's smoother reconstruction.

## 25. Consolidated Results

| Model             | MSE      | MAE      | SSIM     | Parameters | Time (s)  |
|-------------------|----------|----------|----------|------------|-----------|
| FC Autoencoder    | 0.022020 | 0.058438 | 0.737232 | 211040     | 18.646761 |
| Conv. Autoencoder | 0.002623 | 0.015045 | 0.973955 | 74497      | 20.188794 |
| Denoising CAE     | 0.004915 | 0.021914 | 0.942025 | 74497      | 18.930042 |
| VAE               | 0.045195 | 0.107254 | 0.468692 | 284933     | 23.994515 |

## 26. Discussion Questions

- 1. What is an autoencoder?**  
An autoencoder is a type of neural network that learns to reproduce the input it receives. It first compresses the input into a smaller representation and then reconstructs the original image from it.
- 2. What is the purpose of the encoder?**  
The encoder takes the input image and converts it into a smaller representation called the latent representation. It tries to keep the most important information from the image.
- 3. What is the purpose of the decoder?**  
The decoder takes the compressed latent representation and tries to reconstruct the original image as accurately as possible.
- 4. What is a latent representation?**  
A latent representation is a compact version of the input created by the encoder. It contains the important features of the image in a smaller form.
- 5. Why is the latent representation usually smaller than the input?**  
Making it smaller forces the model to learn the important features instead of simply copying the input. It also allows us to compress the data.
- 6. What happens when the latent dimension is too small?**  
If it is too small, the model cannot store enough information about the image. As a result, the reconstructed image may lose important details and have a higher reconstruction error.
- 7. What happens when the latent dimension is increased?**  
A larger latent space can store more information, so the reconstructed images usually become better. However, we get less compression and the model may require more parameters.
- 8. Why can a convolutional autoencoder preserve spatial information more effectively?**  
Convolutional layers look at small local areas of an image. This helps the network understand things like edges, curves, and nearby pixels, which are very important for images.
- 9. Differentiate an autoencoder and a denoising autoencoder.**  
A normal autoencoder tries to reconstruct the same input image. A denoising autoencoder is given a noisy or corrupted image and learns to reconstruct the original clean image.
- 10. Why is the clean image used as the target in denoising?**  
The clean image tells the model what the correct output should look like. This helps the network learn to remove the noise instead of simply reproducing it.
- 11. What happens when the noise level is increased?**  
When there is more noise, the reconstruction becomes more difficult. In our experiment, higher noise resulted in higher MSE and MAE and lower SSIM.
- 12. Why are MSE, MAE and SSIM useful for image reconstruction?**  
They measure reconstruction quality in different ways. MSE measures squared pixel differences, MAE measures the average absolute difference, and SSIM checks how similar the overall structure of the images is.
- 13. What is a Variational Autoencoder?**  
A Variational Autoencoder, or VAE, is an autoencoder that learns a probability distribution for the latent space instead of learning just one fixed latent value for each image.
- 14. How does a VAE differ from a conventional autoencoder?**  
A normal autoencoder produces a fixed latent representation. A VAE learns two values, called mean ( $\mu$ ) and variance ( $\sigma^2$ ), and uses them to sample a latent vector. It also uses KL divergence to keep the latent space well organized.
- 15. Why does a VAE learn a distribution instead of a single latent vector?**  
Why does a VAE learn a distribution instead of a single latent vector? Learning a distribution makes the latent space more continuous and organized. This allows us to sample new points and generate new images.
- 16. What is the reparameterization trick?**  
It expresses sampling as  $z = \mu + \sigma \odot \epsilon$ , where  $\epsilon$  is sampled from a standard normal distribution. This keeps the sampling operation differentiable with respect to the encoder parameters.

17. **Why is  $\epsilon$  sampled from a standard normal distribution?**

It provides a fixed source of random variation while the learned  $\mu$  and  $\sigma$  control the resulting latent distribution.

18. **What is the role of KL divergence in the VAE loss?**

KL divergence encourages the learned latent distribution to stay close to the standard normal distribution. This makes the latent space smoother and allows us to generate new samples more easily.

19. **What is the prior distribution used in the VAE?**

The prior used in this experiment is the standard normal distribution,  $p(z) = \mathcal{N}(0, I)$ .

20. **Why is a two-dimensional latent space useful for visualization?**

A two-dimensional latent space is easy to plot on a graph. We can visually see where different images are located and how they are organized in the latent space

21. **What does latent-space interpolation demonstrate?**

It shows what happens when we gradually move from one latent point to another. If the model has learned a good latent space, the generated images should change smoothly between the two images.

22. **Why can a VAE generate new images?**

A VAE learns a continuous latent space. We can randomly sample a point from the latent distribution and give it to the decoder, which then produces a new image.

23. **Why might some VAE-generated images be unclear?**

The latent space has limited capacity, especially when we use only two dimensions. The VAE also has to balance reconstruction quality with regularization, so some small details may be lost, making the generated images look blurry.

24. **What is the difference between reconstruction and generation?**

Reconstruction means taking an existing image and trying to reproduce it. Generation means sampling a new latent point and using the decoder to create a new image.

25. **Why should test images remain unseen during training?**

Test images should not be used during training so that we can properly check whether the model works on new, unseen data. This gives us a better idea of how well the model generalizes.

## 27. Additional Exercises

1. Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.
2. Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.
3. Train the denoising autoencoder using two different noise levels and compare SSIM.
4. Replace the convolutional decoder with transposed convolutions and compare the reconstructions.
5. Implement a VAE with latent dimension 2 and 8 and compare the latent representations.
6. Investigate the effect of increasing the KL-loss contribution.
7. Generate a larger grid of VAE samples and discuss diversity.
8. Select two digit regions in the latent space and perform interpolation.

## 28. Expected Outcome

At the end of the experiment, the complete image-representation and reconstruction pipeline was demonstrated:

Image  $\rightarrow$  Encoder  $\rightarrow$  Latent Representation  $\rightarrow$  Decoder  $\rightarrow$  Reconstructed Image

The denoising pipeline was also demonstrated:

Clean Image  $\rightarrow$  Noise  $\rightarrow$  Denoising Autoencoder  $\rightarrow$  Recovered Image

and the variational generation pipeline:

Image  $\rightarrow (\mu, \sigma) \rightarrow z \rightarrow$  Decoder  $\rightarrow$  Generated/Reconstructed Image

**Inference:** The completed experiment demonstrates the progression from deterministic compression and reconstruction to spatially aware convolutional reconstruction, denoising under controlled corruption, and probabilistic latent-space generation. The numerical results and visualizations consistently show that convolutional structure is particularly effective for MNIST reconstruction, while the VAE provides a useful continuous latent space for generation and interpolation at the cost of some reconstruction fidelity.

## 29. Github Link

[https://github.com/shaliniparvathareddy/lab7\\_DeepLearning/tree/main](https://github.com/shaliniparvathareddy/lab7_DeepLearning/tree/main)

## 30. References

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